Abdelrahman Elawady

Madinah, Saudi Arabia | Portfolio Website | GitHub | LinkedIn

EDUCATION

University of Prince Muqrin UPM — *Madinah, Saudi Arabia*

- B.Sc. in Electrical Engineering – GPA: 3.99/4 (1st Honor's) — *Sep 2023 – Expected 2027*

EXPERIENCE

Electrical Engineering Intern — *Jun 2025 – Aug 2025*

Jabal Al-Alam LTD Co. – JAA (Delta Controls partner) — *Madinah, Saudi Arabia*

- Assembled and configured 500+ GRMS classic control electrical panels with relays and controllers and inspected them on site.
- Drafted I/O lists in Excel for GRMS and BMS applications, ensuring accurate mapping and efficient field devices testing.
- Programmed building automation controllers using GCL+ structured text and configured database software objects via BACnet, MS/TP/Ethernet/IP and Modbus network protocols in enteliWEB for commercial and institutional buildings.
- Conducted on-site testing, commissioning, and troubleshooting of BMS field devices and panels (MCCs, DDCs, AHUs, Chillers, pumps, sensors, actuators, VFDs, switches) with a primary focus on HVAC and chilled water system control.
- Designed AutoCAD layouts, fabricated laser-cut Nameplates, and created real-time enteliVIZ monitoring interface graphics.

Projects

2-wheel self-balancing robot — *Ongoing*

- Designing a 2-wheel self-balancing robotic platform by integrating an IMU sensor with a PID controller, tuned via Bluetooth.
- Implementing Field-Oriented Control (FOC) for Brushless DC (BLDC) motors using an Arduino-based FOC library.

Automated Hospital Logistics PLC Conveyor System — *Ongoing*

- Designing ladder logic control structures to manage automated material handling and bin staging for a hospital supplies conveyor.
- Developing a predictive supply and demand forecasting AI regression model techniques to optimize routing efficiency.

*Ionic Plasma Thruster project – **1st place in EE All-levels competition*** — *Apr 2025 – May 2025*

- Engineered a fuel-less electric propulsion system using high-voltage corona discharge, strategically selecting cost-effective materials to meet budget constraints, and powered it by 12v DC power supply and lithium batteries.
- Designed and implemented a ZVS soft-switching driver, generating 40kv+ high-voltage AC, and integrated a modular design into a functional 3D-printed sailing boat prototype to successfully demonstrate space-grade technology in a marine application.

Smart IoT LED Display Systems for Campus Offices — *Nov 2024 – Feb 2025*

- Pioneered and designed Bluetooth-enabled LED display systems for the Dean's Office and Career Center using ESP32 microcontrollers, optimizing RAM and DMA allocation to ensure flicker-free rendering on a low-latency UART interface.
- Deployed custom web app and engineered cost-effective solutions that reduced implementation costs by approximately 75% compared to commercial alternatives with low maintenance for professional environments.

Wireless Bluetooth-Controlled 4-Axis Robotic Arm — *Dec 2024 – Jan 2025*

- Collaborated with an engineering team to design and construct a 4-axis 3D printed robotic arm for precise pick-and-place operations, utilizing an Arduino to process motion control inputs from a custom potentiometer Bluetooth interface.

*Electric generator project – **2nd place in Physics competition*** — *Apr 2024 – May 2024*

- Designed and constructed a low-budget axial flux electric generator in a trial-and-error approach utilizing 250-loop coils and magnets mounted on a disk, successfully generating a continuous 3V output to validate electromagnetic induction principles.

Publications

- **Elawady, A.**, Mohamed, J. A., & Bin Abid, M. (2025). Advancements in microplastics detection techniques and their multidimensional impacts on aquatic ecosystems and human health. *Journal of Hazardous Materials: Plastics* (**Q1**).
- **Elawady, A.,** et al. (2025). Polymer-Based Innovations in Concrete: A Review on Enhancing Self-Healing Capabilities. *Innovative Infrastructure Solutions* (**Q2**).
- Bin Abid, M., **Elawady, A.**, & Shamim, R. (2025). Exploring AlUla, Saudi Arabia: a journey through history, heritage, resources, and tourism. *GeoJournal* (**Q2**).

SKILLS & TECHNOLOGIES

Engineering Tools: AutoCAD, TIA portal, Simulink, Multisim, KiCad, ETAB, Proteus, MS office (Word, Excel, PowerPoint).
Software/ Programming: Siemens PLC Ladder Logic, Assembly, C/C++, Python, AI PyTorch, MATLAB, HTML, JavaScript, Git.
Hardware: Microcontrollers (ESP32, Arduino), PCB design, soldering, Electronics & IoT, 3D printing, Oscilloscope, Multimeter.

LEADERSHIP & AWARDS

Electrical Engineering club President – UPM — Feb 2025 – Sep 2025

- Spearheaded the Electrical Engineering Club, coordinating workshops and field trips & organizing department-wide competitions.

Prince Faisal bin Salman Award for academic excellence – UPM — 2025

- Recipient of His Royal Highness Prince Faisal bin Salman Award for academic excellence, earning a 10,000 SAR merit prize.